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Author(s): Strohmann, Timo; Schoormann, Thorsten; Khosrawi-Rad, Bijan; Siemon, Dominik; Tuunanen, Tuure

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Synthesizing Design Knowledge for Persuasive Conversational Agents in Sustainability

Timo Strohmann
Technische Universität Braunschweig, t.strohmann@tu-bs.de

Thorsten Schoormann
Roskilde University, tschoormann@ruc.dk

Bijan Khosrawi-Rad
Technische Universität Braunschweig, b.khosrawi-rad@tu-bs.de

Dominik Siemon
LUT University, dominik.siemon@lut.fi

Tuure Tuunanen
University of Jyväskylä, tuure@tuunanen.fi

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Synthesizing Design Knowledge for Persuasive Conversational Agents in Sustainability

Special Interest Group on Green Information Systems
SIGGreen Pre-ICIS 2024 Full Workshop Paper

Timo Strohmann
Technische Universität Braunschweig
t.strohmann@tu-bs.de

Thorsten Schoormann
Roskilde University
tschoormann@ruc.dk

Bijan Khosrawi-Rad
Technische Universität Braunschweig
b.khosrawi-rad@tu-bs.de

Dominik Siemon
LUT University
dominik.siemon@lut.fi

Tuure Tuunanen
University of Jyväskylä
tuure@tuunanen.fi

Abstract

Global sustainability challenges depend on large-scale behavioral change. Persuasive conversational agents (CAs) have emerged as promising systems to shape user attitudes and foster sustainable actions. However, the design knowledge surrounding these systems is scattered and seldom reused. This paper follows a meta-synthesis approach and consolidates insights from 26 publications directly addressing persuasive CAs and sustainability. We extracted and clustered over 500 design knowledge chunks into 30 abstract design clusters, which are organized in a three-dimensional framework: Sustainability by agent (how CAs foster sustainability goals), sustainability in agent (how CAs embed ethical and inclusive practices), and conversational agent (core CA features for naturalistic interaction and adaptability). Our findings illuminate current design practices, uncover blind spots, and lay the groundwork for advanced theorizing. This structured resource enables researchers and practitioners to systematically identify important chunks to design, implement, and evaluate persuasive CAs that reinforce sustainable behaviors while maintaining responsible practices. By synthesizing fragmented insights into a reusable knowledge base, we promote consistency in future design efforts and offer a firm foundation for developing an overarching design theory of CA-based interventions for sustainability.

Keywords

Sustainability, meta-synthesis, conversational agents, persuasiveness, behavior change support system, design knowledge, design principles, design theory.

Motivation

Global sustainability challenges require significant shifts in human behavior (Melville 2010; United Nations 2015). Achieving the UN's Sustainable Development Goals (SDGs) relies on individual and collective actions toward behavior changes and more sustainable practices which can be nudged by technological innovations (Henkel and Kranz 2018; Ixmeier and Kranz 2024; Melville 2010). Persuasive systems aim to change users' attitudes or behaviors without coercion (Fogg 2003). Conversational agents (CAs) have emerged as promising technological solutions due to their ability to interact using natural language and their perceived humanness, which enhances social presence (Araujo 2018). These human-like qualities set them apart from

traditional persuasive systems and contribute to their effectiveness by engaging users on a social level (Nass and Moon 2000). Advancements in artificial intelligence, especially in natural language processing and large language models like OpenAI's GPT (Radford et al. 2019), have significantly enhanced CAs' capabilities, making them promising tools for influencing behavior and potentially promoting sustainable practices. Recent studies have explored how CAs can raise awareness, share information, and influence user behaviors toward sustainability goals (Diederich et al. 2019; Hillebrand and Johannsen 2021; Müller and Reuter-Oppermann 2024). For instance, a chatbot to encourage climate-friendly behavior among employees (Hillebrand and Johannsen 2021), a persuasive CA to promote sustainable mobility beliefs for e-bike usage (Diederich et al. 2019), and a behavior change support system as a CA to increase blood donation behavior (Müller and Reuter-Oppermann 2024). These examples illustrate the effectiveness of CAs in shaping beliefs and actions toward sustainability by leveraging their human-like interaction capabilities, as explained by theoretical mechanisms like the computers are social actors (CASA) paradigm (Melville 2010; Nass and Moon 2000).

With these developments, we observe an increasing number of artifacts and design knowledge chunks (vom Brocke et al. 2020) to purposefully craft this class of systems, such as in the form of design principles (DPs) (Gregor et al. 2020). Since the design knowledge is scattered across different application domains, there is a need to systematically analyze and consolidate it for the sustainability context. This would help to increase the reusability of existing insights and advance rigor (Elshan et al. 2022; Khosrawi-Rad, Grogorick, et al. 2024; Schoormann et al. 2025). Therefore, synthesizing existing design knowledge is essential to enhance its practical impact. This paper addresses the fragmentation of design knowledge for sustainability-focused CAs in the IS domain by asking three research questions (RQ):

- **RQ1**: What is the status quo of design knowledge for persuasive CAs in sustainability, in promoting sustainability within the IS domain?
- **RQ2**: How can existing insights be systematically synthesized into a reusable, structured resource that informs the design of persuasive CAs in sustainability?
- **RQ3**: In what ways can this synthesized knowledge guide future research directions and the development of an overarching design theory?

In pursuing to answer these, we conducted a meta-synthesis of 66 relevant publications. Through iterative analysis and clustering of over 500 design knowledge chunks extracted from our corpus, we present a framework illustrating how CAs can effectively promote sustainable behaviors and embody sustainability principles. We consolidate scattered design insights into a three-dimensional framework with components for 'sustainability by agent' (i.e., how CAs foster sustainability goals), 'sustainability in agent' (i.e., how CAs adhere to ethical, inclusive practices), and 'conversational agent' (i.e., general and core CA features). Our work contributes to theoretical advancement by mapping isolated design principles onto a unified resource, thus enhancing reusability and providing a basis for deeper theorizing. In practice, it offers structured guidance for researchers and practitioners who seek to develop and evaluate persuasive CAs in sustainability that effectively drive sustainable behavior change.

Research Background

Sustainable Development

Ensuring fair and livable conditions for future generations has long been recognized as crucial (Brundtland 1987). Widely accepted frameworks—such as the Triple Bottom Line (Elkington 1997) and the 17 SDGs (United Nations 2015)—offer ways to operationalize sustainability. In the IS field, research on Green IT (Loos et al. 2011) and Green IS (vom Brocke et al. 2013) demonstrates how technology can be designed, used, and disposed of more responsibly. Building on this, our paper focuses on two perspectives: designing conversational agents sustainably ("in agent") and using them to foster sustainable behavior ("by agent"), reflecting the broader notion of digital sustainability (Kotlarsky et al. 2023).

Conversational Agents

CAs are systems that interact via natural language, both written and spoken, evolving from early programs like ELIZA (Weizenbaum 1966) to today's LLM-based solutions such as GPT-4 (Teubner et al. 2023) and

mimic human-to-human conversations (Gnewuch et al. 2017). By integrating persuasive design and social cues (Nass and Moon 2000), CAs can build trust, display empathy, and offer personalized, context-aware guidance (Strohmann et al. 2022), all of which are vital for promoting sustainable behavior. The CA landscape includes task-oriented chatbots (McTear 2017), virtual assistants (Gnewuch et al. 2017), and virtual companions (Strohmann et al. 2022). With the growing attention, we also observe a growing landscape of design knowledge. Despite abundant design insights, a unified framework for sustainability-oriented CA design is still lacking, underscoring the need for a systematic synthesis of existing knowledge.

Design Knowledge in Design Science Research

Design science research (DSR) produces design artifacts as well as actionable design knowledge that enables their users to instantiate an artifact of a similar class (Baskerville et al. 2018; Simon 1996). Generally, design knowledge follows a prescriptive nature and explains how an artifact could be designed to achieve a particular outcome. The designer generates design knowledge that must be accumulated to advance the knowledge base. Codification helps to store, elevate, and make chunks of design knowledge reusable that emerge during the building and evaluation of artifacts (Chandra et al. 2015) – design knowledge chunks represent a reusable lesson learned from a design project (vom Brocke et al. 2020). Thereby, the form of codified knowledge can differ (Möller et al. 2022). Common formats and concepts related to design knowledge include meta-requirements and meta design informed by kernel theories from social or natural science (Walls et al. 1992), design principles (Gregor et al. 2020), and design features that are more closely related to the situational implementation (Schoormann et al. 2024). This knowledge is important to make sure not to repeat similar failures during a project and build upon prior successful designs.

Methodology

We are applying a six-step framework for meta-synthesis on design knowledge (Khosrawi-Rad, Grogorick, et al. 2024) to analyze DPs related to persuasive systems, behavior change support systems, and CAs in the context of sustainability in IS (see Figure 1). Khosrawi-Rad et al.'s (2024) framework is a method to analyze design knowledge according to DSR-specific criteria, to identify and discuss contradictions of design knowledge, and to synthesize the knowledge base into abstract DP sets.

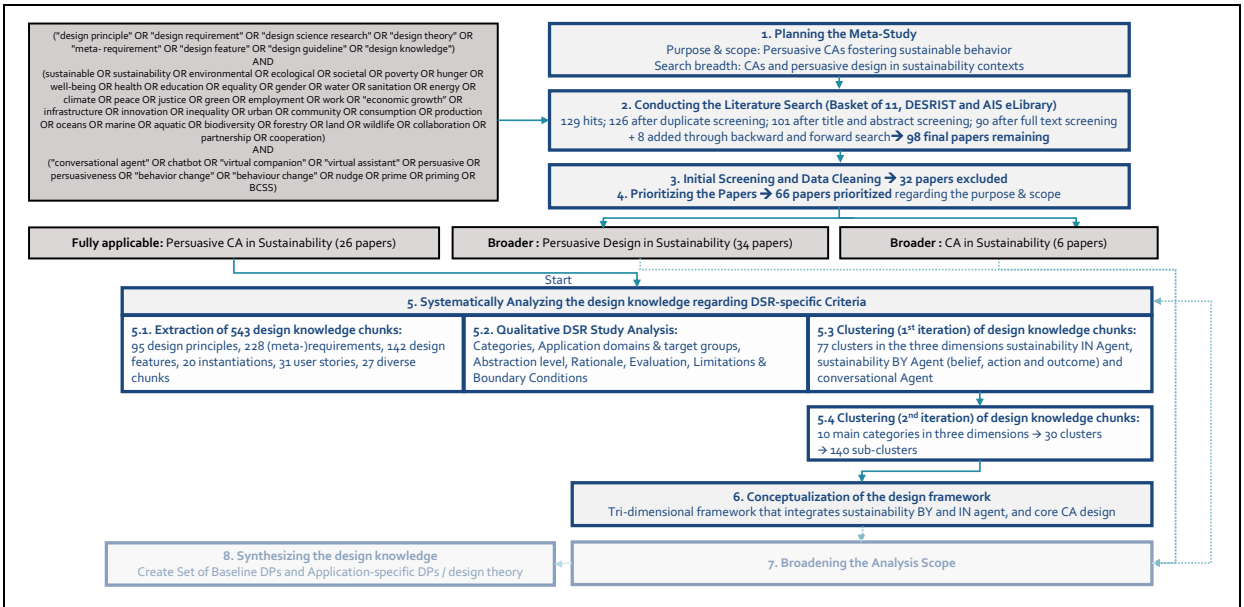


Figure 1. Meta-Synthesis Procedure according to Khosrawi-Rad et al. (2024)

In **step 1**, we defined 'design knowledge for persuasive CAs in sustainability' as our overall purpose and scope. In **step 2**, we conducted a literature search in AIS eLibrary, the Basket of 11 IS journals, and

DESRIST proceedings, combining synonyms for design knowledge, sustainability (including SDG-related terms), and CAs or persuasiveness. From an initial 129 results, we removed duplicates and screened for relevance, resulting in 98 final papers. In **step 3**, we ensured that each publication offered design knowledge aligned with our purpose, discarding 32 papers. In **step 4**, we prioritized the remaining 66 papers by differentiating three scopes: fully applicable (26 papers) focusing on persuasive CAs in sustainability, broader persuasive design in sustainability (34 papers), and broader CAs in sustainability (6 papers). This allowed us to highlight the most relevant subset for in-depth analysis, while still retaining insight into a broader range of studies. We initially focused on the 26 fully applicable papers for a deeper analysis due to their clear emphasis on persuasive CAs in sustainability, while we will examine the remaining 40 papers subsequently for content saturation and further design insights.

In **step 5**, we systematically extracted design knowledge chunks (vom Brocke et al. 2020) from each paper, including design principles, (meta-)requirements, features, instantiations, and other prescriptive insights (*step 5.1*). We systematically coded each publication and conducted a qualitative assessment, discussing possible cluster groupings and defining an initial framework, laying the groundwork for the more detailed analysis and refinement of the clustering in the subsequent steps (*step 5.2*). We then conducted the next two iterations of clustering to organize the design knowledge chunks. In the first iteration (*step 5.3*), we formed 77 clusters around three main dimensions: sustainability in the agent, sustainability by the agent (covering belief, action, and outcome, based on the BAO framework (Melville 2010)), and conversational agent in general. In the second iteration (*step 5.4*), the first author refined these initial groupings to arrive at 10 main categories, which further led to 30 clusters and 140 sub-clusters and again assigned those to all 543 design knowledge chunks; the cluster names within each dimension were iteratively discussed with the other authors to ensure coherence and consensus. We also analyzed each paper according to relevant DSR criteria, such as kernel theories used, evaluation methods, application domain, SDG context, and the social, environmental, or economic focus of sustainability. Additionally, we captured any available instantiations of the described artifacts and documented all this information and design knowledge in a multi-table database in *Airtable*[1]. Separate tables were created for studies, design knowledge chunks, both clustering iterations, and kernel theories, and these were cross-linked to enable future retrieval and analysis. Four researchers were involved in the overall coding, extraction, and analysis, with the first author taking the lead role in managing the database and guiding the clustering process, and consistently discussing the developing structure and categories with the others. In **step 6**, we moved on to conceptualizing a design framework based on the refined clusters, which focuses on how persuasive design elements for sustainability can be integrated into CAs. This framework includes both how sustainability principles can be embodied in the agent itself and how the agent can support sustainability goals by influencing user beliefs and actions.

Steps 7 and 8, in which we plan to broaden our analysis scope and synthesize the clusters into higher-order design knowledge, are part of our future work. Since this is a workshop paper, we concentrate here on reporting the methodology and results obtained up to step 6, laying the groundwork for the next tasks.

Analysis of Papers

Following the meta-synthesis procedure, we identified a refined corpus of 26 relevant papers, specifically on persuasive CAs in sustainability. In terms of **publication outlets**, out of 26 studies in the most important set, ICIS was the most common venue with 8 contributions, followed by DESRIST (6), HICSS (3), and PACIS, AMCIS, and ECIS (2). Only two journal outlets were identified, namely BISE and CAIS.

In what follows, we present our preliminary observations based on the extracted design knowledge and highlight emerging patterns across the existing research base. Moreover, regarding contributions toward various sustainability topics, we analyzed all 66 studies. Figure 1 presents two visualizations illustrating how the papers (all 66) address the United Nations **SDGs** and the three **sustainability dimensions** (economic, social, and environmental). The bar chart on the left shows the count of studies mapped to individual SDGs; for example, a noticeable cluster targets SDG 3 (good health and well-being) and SDG 4 (quality education), indicating a particularly high interest in health and educational challenges. Other frequently addressed goals include SDG 13 (climate action) and SDG 12 (responsible consumption and

[1] Airtable is a cloud-based platform that combines spreadsheet functionality with a relational database.

production), whereas several goals—such as SDG 1 (no poverty) and SDG 2 (zero hunger)—are not represented in our sample. The pie chart on the right groups the studies into sustainability dimensions, revealing a strong emphasis on social issues (e.g., well-being, education, equality), followed by environmental initiatives (e.g., reducing carbon emissions). Overall, these graphs underscore a predominantly social and environmental orientation in the literature on sustainability-related CAs.

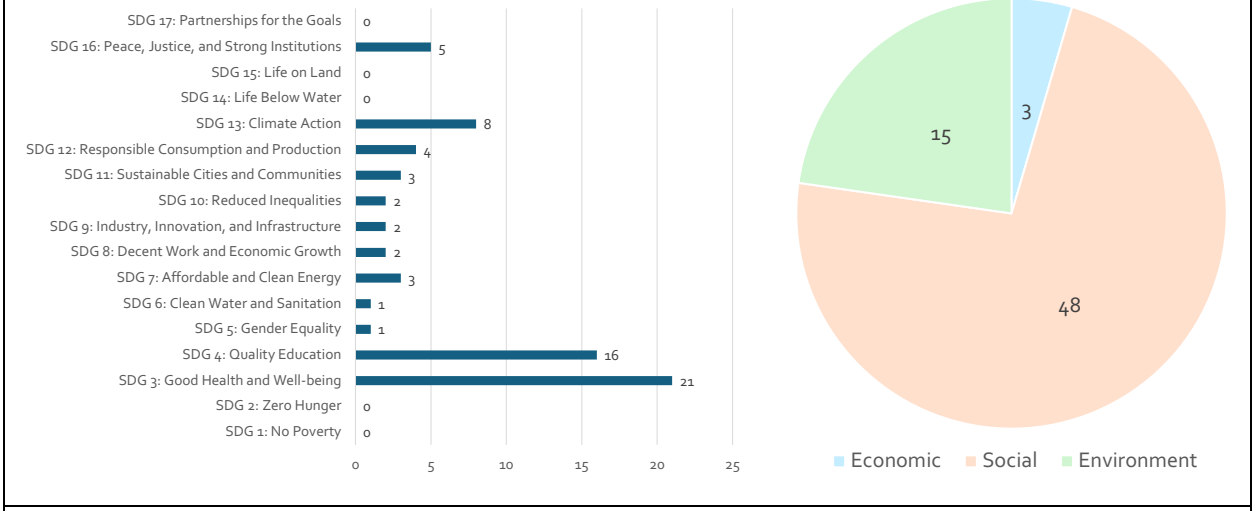


Figure 1: Sustainability Distribution of Publications (n=66 papers)

We identified 41 **kernel theories** and **justificatory knowledge elements** that informed the design and evaluation of persuasive CAs in sustainability contexts. Table 1 highlights seven widely used theories—each employed by at least two studies and not confined to a single domain (like learning). These theories encompass behavior change and motivation, as well as CA-specific perspectives, to guide the design and evaluation of persuasive conversational agents in sustainability contexts.

Area	Theory name	Key idea	Example paper(s)
Behavior change and motivation	**Self-Determination Theory** (Deci and Ryan 1985)	Explains how autonomy, competence, and relatedness drive intrinsic/extrinsic motivation.	Müller & Reuter-Oppermann (2024); Schlimbach et al. (2024); Khosrawi-Rad et al. (2024); Strunk et al. (2024)
	Theory of Planned Behavior (Ajzen 1991)	Attitudes, subjective norms, and perceived behavioral control predict intention and behavior.	Hillebrand & Johannsen (2021); Parmar et al. (2009); Diederich et al. (2019)
	Transtheoretical Model (Prochaska and DiClemente 1982)	Behavior change is a cyclical process of pre-contemplation, contemplation, preparation, action, and maintenance.	Müller & Reuter-Oppermann (2024); Meywirth (2024)
	Behavior Change Theory (Michie et al. 2011)	Focuses on interventions that target capability, opportunity, and motivation to shape behavior.	Aljaroodi et al. (2017); Breiter et al. (2024)
	Persuasive Systems Design (Oinas-Kukkonen and Harjumaa 2009)	Provides a process and feature categories (primary task, dialogue, credibility, social support) to shape user attitudes and behaviors.	Meywirth (2024); Weimann et al. (2022); Strunk et al. (2024)
CA-specific	**Computers Are Social Actors** (Nass et al. 1994)	People apply social norms to computers and treat them like social entities.	Schlimbach et al. (2024); Gnewuch et al. (2018); Weimann et al. (2022); Schlimbach, Windolf, et al. (2023); Diederich et al. (2019)
	Uncanny Valley (Mori 1970)	Explains negative reactions to almost-but-not-quite-human artifacts, highlighting design trade-offs in human-like interfaces.	Weimann et al. (2022); Schlimbach, Windolf, et al. (2023)

Table 1. Relevant Kernel Theories and Justificatory Knowledge

In our synthesis of 26 papers, we extracted a total of 543 distinct **design knowledge chunks**. The most prominent were 95 DPs contributed across 22 papers, 228 (meta-)requirements across 20 papers, 142 design features in 8 papers, 20 instantiations from 18 papers, and 31 user stories in 5 papers. The remaining elements comprised more specific types of knowledge, such as frameworks or models.

Table 2 illustrates how individual design knowledge chunks from a single study (Müller and Reuter-Oppermann 2024) are systematically assigned to one or more sub-clusters, clusters, and design categories. Each row represents a distinct chunk (e.g., DF6.1 as a design feature and DP1 as a design principle). Design features typically address one specific design aspect—here, "appointment reminders." In contrast, design principles tend to span multiple sub-clusters because they often describe more comprehensive guidelines or interventions that integrate several design aspects. In this example, DP1 covers reminders and notifications but also extends to organizational planning support and proactive capabilities, showing how a single principle can intersect with multiple clusters.

Chunk ID	Type	Source	Sub cluster	Cluster	Design category
DF6.1	DF	"Appointment reminder"	BY3.2.1. Reminders	BY3.2. Proactive & Trigger-Based Nudges	BY3. Recommendations & Nudging
DP1	DP	"Provide the BDBCSS with proactive planning interventions comprising reminders as well as push notifications and with organizational and planning assistance to enable users to integrate and prioritize blood donations in their busy lifestyles."	BY3.2.1. Reminders	BY3.2. Proactive & Trigger-Based Nudges	BY3. Recommendations & Nudging
			BY3.2.2. Notifications & Alerts		
			BY3.3.1. Digital Scheduling	BY3.3. Organizational & Planning Support	
			CA2.1.2. Proactivity	CA2.1. Context-Awareness & Proactivity	CA2. Context and Personalization

Table 2. Illustrative Clustering of Design Knowledge Chunks from Müller and Reuter-Oppermann (2024)

Table 3 provides a concise concept matrix of our 26 papers, highlighting both their sustainability focus (social or environmental – economic is not present), the specific SDGs they address, and whether they include a mapping diagram of design knowledge. It also shows the types of design knowledge each study contributes, as well as which design categories (BY1–BY5, IN1–IN2, and CA1–CA3) they cover. "BY" clusters capture how conversational agents promote sustainability by fostering awareness & education (BY1), motivation & gamification (BY2), recommendations & nudging (BY3), collaboration & social networking (BY4), feedback & reflection (BY5). The "IN" clusters address sustainability in the agent through ethical and responsible design (IN1) and usability and accessibility (IN2). Finally, the CA clusters focus on general CA aspects such as interaction and communication (CA1), context and personalization (CA2), and system and implementation (CA3). An x in a cell indicates that the respective concept is fulfilled.

Figure 2 presents the 30 final design clusters that emerged from our iterative clustering process. Each cluster is grouped under a broader category (BY1–BY5, IN1–IN2, CA1–CA3) and is further refined by sub-clusters (e.g., BY1.1, BY1.2, BY1.3), which capture more specific design focuses. These 30 clusters are further broken down into 140 sub-clusters, providing fine-grained detail on how to design persuasive CAs in sustainability. The full breakdown of clusters and sub-clusters is provided in the digital appendix (https://doi.org/10.6084/m9.figshare.28389158). These clusters collectively represent how sustainability-oriented design knowledge—ranging from awareness-raising mechanisms (BY1) to ethical considerations (IN1) and system implementation aspects (CA3)—can be systematically organized to guide the creation of persuasive conversational agents. Moreover, the digital appendix includes an overview of the analyzed studies, all kernel theories, as well as a description for each cluster.

	Sustainability dimension			Design knowledge type and construction						Design category									
Paper reference	**Social**	**Environment**	**SDG**	**Mapping diagram**	**Design principle**	**Meta-requirement**	**Design feature**	**Instantiation**	**User story**	**BY1. Awareness & Education**	**BY2. Motivation & Gamification**	**BY3. Recommen-dations & Nudging**	**BY4. Collaboration & Social Networking**	**BY5. Feedback & Reflection**	**IN1. Ethical & Responsible Design**	**IN2. Usability & Accessibility**	**CA1. Interaction & Communication**	**CA2. Context & Personalization**	**CA3. System & Implementation**
Aldkheel and Zhou (2023)	x		3,10, 16	x	X	x				x	x	x			x	x	x	x	x
Aljaroodi et al. (2017)	x		3		x	x				x	x		x		x	x	x		
Breiter et al. (2024)		x	12	x	x	x				x	x	x			x		x	x	
Diederich et al. (2019)		x	13				x	x			x			x			x	x	
Gnewuch et al. (2018)		x	7		x			x		x		x		x		x			
Hillebrand and Johannsen (2021)		x	13			x		x		x	x	x	x	x	x		x	x	
Hobert (2019)	x		4	x	x	x		x	x	x				x			x		
Khosrawi-Rad et al. (2024)	x		4	x			x			x	x					x			
Lenk and Schmidt (2022)	x		4	x	x	x		x	x			x	x	x			x	x	x
Meier et al. (2019)	x		3	x	x	x				x		x			x	x	x	x	
Meywirth (2024)	x		3	x	x	x		x		x		x					x		
Mulgund et al. (2021)	x		3		x			x				x	x		x	x	x		
Müller and Reuter-Oppermann (2024)	x		3	x	x	x	x	x		x	x	x	x	x			x	x	x
Schlimbach and Khosrawi-Rad (2022)	x		4			x	x	x							x	x	x		x
Schlimbach, Lange, et al. (2023)	x		4	x	x		x			x	x	x	x	x	x		x	x	
Schlimbach, Windolf, et al. (2023)	x		4	x	x	x		x		x	x	x	x	x			x	x	
Schlimbach, Behne, et al. (2023)	x		4	x			x	x		x	x	x	x	x				x	
Schlimbach et al. (2024)	x		4		x	x				x	x	x	x	x	x	x	x	x	x
Stattkus et al. (2024)	x		5	x	x	x		x		x	x		x		x	x	x		
Strunk et al. (2024)		x	11	x	x	x		x			x			x					
Tavanapour et al. (2019)	x		16		x	x				x	x	x	x	x	x	x	x	x	x
Tolzin et al. (2023)	x		4		x	x		x		x				x	x	x	x	x	
Weber et al. (2023)	x		4	x	x	x	x	x	x	x				x				x	
Weimann et al. (2022)	x		3	x	x	x		x			x		x	x		x			
Zahn and Fritsch (2024)	x		4	x	x	x	x	x	x	x	x	x		x			x	x	
Zahn and Schöbel (2024)	x		4	x	x	x		x	x	x	x	x		x		x	x	x	

Table 3. Concept Matrix of the 26 Relevant Papers

Conceptualization of a Design Framework

Having completed our clustering and analysis, we now introduce a consolidated design framework that synthesizes the extracted design knowledge for persuasive CAs in sustainability contexts. The framework features three dimensions: Sustainability by agent, sustainability in agent, and conversational agent. As shown in Figure 2 (see also Table 2 above), the framework is divided into: (1) Sustainability by agent (BY) – How the agent actively supports users' sustainability goals (grounded in Melville's (2010) belief–action–outcome (BAO) framework). (2) Sustainability in agent (IN) – How the agent itself is designed and implemented in a responsible, ethical, and inclusive manner. (3) Conversational agent (CA) – General CA-specific design considerations such as system interfaces, personalization, and human-like communication, which act as an enabler for the other dimensions.

Each dimension contains multiple design clusters. Together, they inform how to craft CAs that are both persuasive toward sustainable behaviors and adhere to sustainability principles. In line with Melville's BAO framework, the sustainability by agent pillar especially leverages a process-oriented logic: it fosters *belief formation* (e.g., awareness, self-efficacy), triggers *actions* (e.g., through gamification, recommendations, or collaboration), and supports *outcome* reflection (e.g., progress tracking and feedback). Crucially, this approach is cyclical: insights from reflection feed back into new awareness or intensified motivation, thus continually reinforcing sustainable attitudes and behaviors. In practice, the framework helps designers decide which design clusters to integrate for their specific application contexts, ensuring a consistent and theory-based approach. By covering both persuasive mechanisms (sustainability by agent), responsible design principles (sustainability in agent), and core CA features, it addresses the critical facets needed for credible, user-oriented, and ethical solutions.

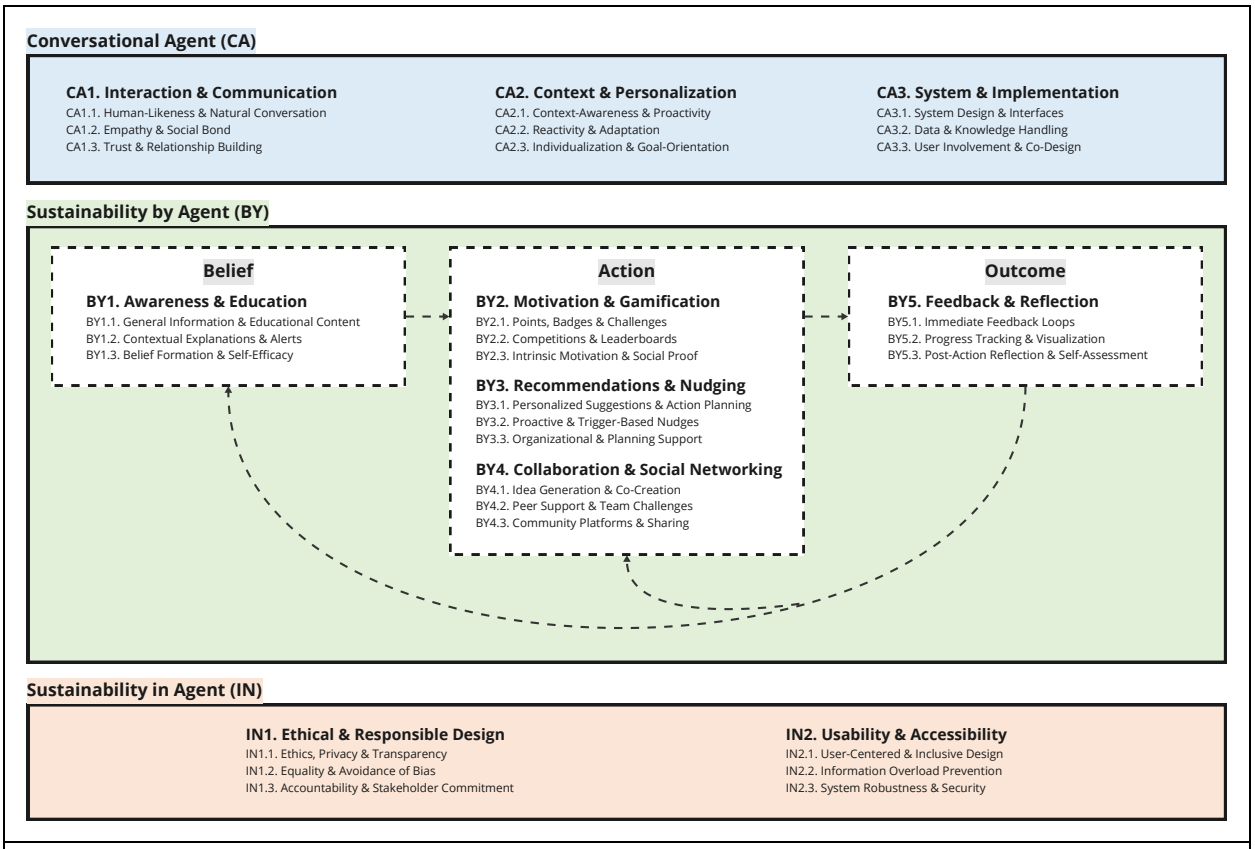


Figure 2: Design Framework for Persuasive Conversational Agents in Sustainability

Sustainability by agent (BY) captures how a **CA** directly contributes to the user's sustainable behaviors. Drawing on Melville's BAO logic, it helps guide the agent's features and interactions so that users (1) form and refine beliefs about sustainability, (2) translate these beliefs into concrete actions, and (3) experience and evaluate outcomes that reinforce (or reshape) subsequent beliefs. Our framework organizes these facets into five main clusters: *BY1. Awareness & education* revolves around providing accessible sustainability information, clarifying why certain habits matter, and helping users form meaningful convictions about their ability to drive change. *BY2. Motivation & gamification, BY3. Recommendations & nudging, and BY4. Collaboration & social networking* collectively focus on transforming belief into actual behavior by providing rewarding strategies such as points, challenges, social proof, real-time reminders, and community support. BY5. *Feedback & reflection* then ensures that users see measurable results from their efforts—through immediate reactions, visualization of progress, and opportunities to reflect and self-assess—and that these new insights feed back into fresh beliefs and future actions.

Sustainability in agent (IN) concerns how the CA itself is conceived and engineered in an ethically sound, inclusive, and robust way. Rather than focusing on how to drive user behavior, this dimension addresses the agent's own adherence to sustainability principles. By safeguarding ethical practices and inclusive usage, sustainability in agent underscores the principle that to promote sustainability, the agent must embody sustainable, respectful practices from the start. *IN1. Ethical & responsible design* tackles data privacy, transparency, the avoidance of biases, and stakeholder accountability, ensuring that the system neither exploits nor misleads users. *IN2. Usability & accessibility* covers inclusive design principles, protection against information overload, and the robust technical underpinnings required to serve diverse user groups fairly and securely.

Finally, **conversational agent (CA)** addresses design facets that are generally relevant to CAs—namely, how to build naturalistic, context-aware, and well-engineered systems. While not unique to sustainability, these aspects are crucial enablers of the agent's persuasive potential. *CA1. Interaction & communication* addresses the style and flow of interaction; designing for human-likeness, empathy, and trust helps build rapport and credibility. *CA2. Context & personalization* ensures the agent adapts to individual user needs, with proactive and reactive capabilities that adjust prompts and guidance as circumstances change. *CA3. System & implementation* focuses on integrating the CA with other technologies, managing data and knowledge in a reliable manner, and involving end users in ongoing development. This tri-dimensional framework provides a structured perspective on how to conceptualize and build persuasive CAs in sustainability. By linking each design feature to a specific part of the framework, researchers and practitioners can systematically incorporate existing design knowledge while ensuring that sustainability by the agent and sustainability in the agent are jointly realized.

Discussion and Future Research

By completing steps three through six, we anticipate achieving a synthesis of design knowledge related to persuasive design and CAs in the context of sustainability within the IS domain. We aim to compile application-specific DPs tailored to sustainability, facilitating the development of customized IS solutions that address specific sustainability challenges. This will involve developing meta-sets of DPs for persuasive design and for CAs in sustainability, thereby enhancing the reusability and practical applicability of design knowledge in these areas. To further support this effort, we plan to make our underlying database of design knowledge chunks publicly accessible, enabling practitioners to filter for suitable design implications and tailor them to their unique sustainability goals.

In summary, we make a threefold **contribution** with our work: First, we provide a synthesized overview of design knowledge at the intersection of CAs and sustainability. This synthesis provides practitioners with concrete design knowledge for developing persuasive CAs that address diverse sustainability challenges through user behavior change, while also contributing to the academic knowledge base by compiling core kernel theories relevant to these systems. Second, our overview systematically reveals underexplored areas in the literature by identifying specific gaps, such as the limited focus on certain SDGs (e.g., SDG 1 and SDG 2), the underutilization of methods like longitudinal field studies and user-centered evaluations, and the potential for incorporating additional kernel theories and context-specific factors into design. Third, our work lays the groundwork for advancing our synthesized design framework into a design theory of persuasive CAs for sustainable behavior. Building on our three-dimensional framework, we advance our theorizing through a multi-faceted reasoning approach. Specifically, we employed inductive reasoning by aggregating and clustering specific design knowledge from the literature to generate our framework. In parallel, we used deductive reasoning to integrate established constructs, such as the SDGs and identified kernel theories relevant to persuasive CAs in sustainability. Building on these foundations, we will proceed with abductive reasoning, a form of logical inference that seeks the simplest and most likely conclusion from our observations. Through this abductive process, we aim to derive a design theory as the most plausible explanation emerging from our synthesized data. For theory construct, it is also needed to establish construct clarity and define precise boundary conditions, delineating the scope of the design theory with attention to validity, reliability, and applicability across individual, organizational, and societal levels. In addition, preliminary validation efforts are planned to evaluate the theory's internal consistency

and external applicability, providing an initial form of proof for its robustness. Moreover, our approach emphasizes the originality and value of the theory by explicitly defining and logically connecting the relationships between constructs with theoretical precision and rigor. This integrated methodology provides a solid foundation for future research and practical implementations aimed at promoting sustainable behavior through persuasive CAs.

As with every research, our research is not free of **limitations**. First, we report only on the core sample collected (i.e., 26 papers). In the next steps, we engage with the broader sample of 66 papers to complement and/or validate our existing design insights. This will increase the robustness and comprehensiveness of our work. Second, to have a manageable set of elements, we initially focused on selected types of design knowledge which can be extended by focusing also on, for instance, design decisions or instantiations provided in a paper. Third, our framework and design knowledge synthesis are justified based on existing work. While the studies themselves often have evaluated their knowledge pieces, evaluating our aggregated design knowledge is part of our future work. Overall, our study brings together a wide array of previously scattered design knowledge on how to build and deploy persuasive CAs in sustainability. By consolidating this knowledge into a single framework, we have taken a crucial step toward an overarching design theory that can guide both scholarly inquiry and real-world implementations.

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